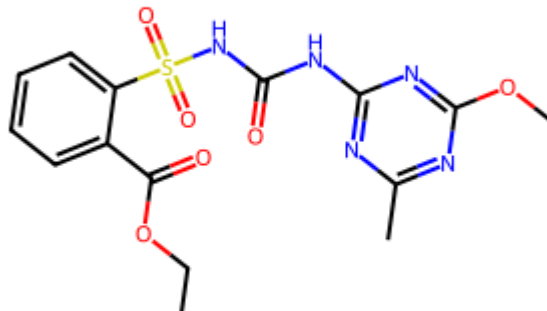


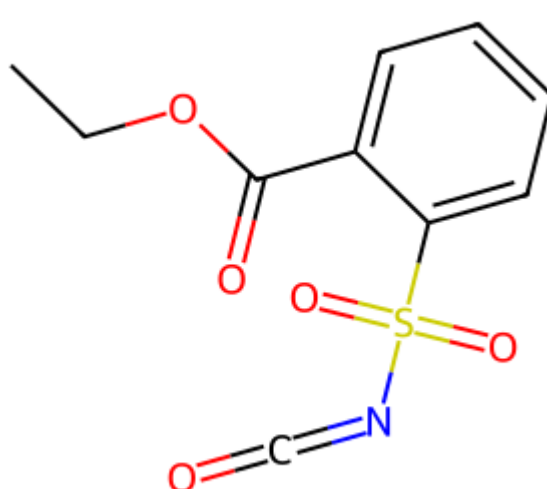
{'analysis_time': 0.0,
'first_solution_time': 0.192755,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.030232699128965687,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 0.030232699128965687,
'retro_star_value': 0.030232699128965687}

Reaction
{'analysis_time': 0.192755,
'first_solution_time': 0.605449,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 8.728160721225827,
'retro_star_reaction_number': 8.728160721225827,
'retro_star_rxn_cost': 4.347192501872942,
'retro_star_value': 8.728160721225827}

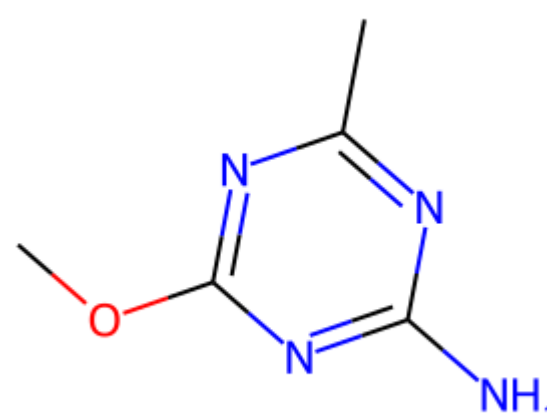


{'analysis_time': 0.192755,
'first_solution_time': 0.605449,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 4.380968219352884,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 4.380968219352884,
'retro_star_value': 8.728160721225827}

Reaction
{'analysis_time': 0.503198,
'first_solution_time': 0.605449,
'num_calls_rxn_model': 4,
'num_calls_value_function': 109,
'retro_star_min_cost': 9.52020440301785,
'retro_star_reaction_number': 9.52020440301785,
'retro_star_rxn_cost': 0.7840270717357756,
'retro_star_value': 13.867396904890793}

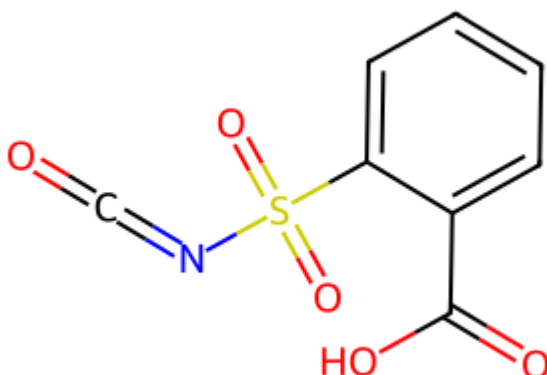


{'analysis_time': 0.503198,
'first_solution_time': 0.605449,
'num_calls_rxn_model': 4,
'num_calls_value_function': 109,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 8.736177331282075,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 8.736177331282075,
'retro_star_value': 13.867396904890793}

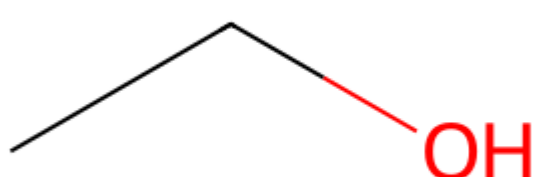


{'analysis_time': 0.503198,
'first_solution_time': 0.503198,
'num_calls_rxn_model': 4,
'num_calls_value_function': 109,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 13.867396904890793}

Reaction
{'analysis_time': 0.601489,
'first_solution_time': 0.950405,
'num_calls_rxn_model': 5,
'num_calls_value_function': 158,
'retro_star_min_cost': 12.467455372856495,
'retro_star_reaction_number': 12.467455372856495,
'retro_star_rxn_cost': 1.3619449304651041,
'retro_star_value': 17.598674946465213}

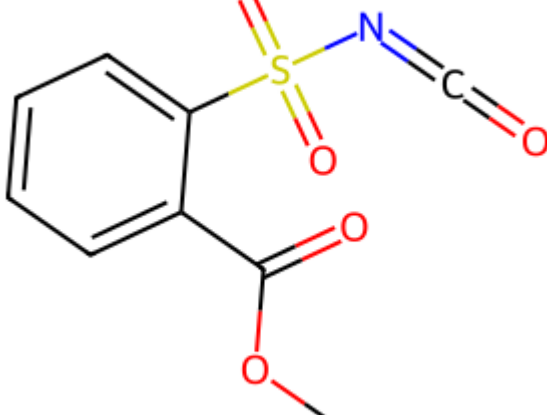


{'analysis_time': 0.601489,
'first_solution_time': 0.950405,
'num_calls_rxn_model': 5,
'num_calls_value_function': 158,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 10.656801347194646,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 10.656801347194646,
'retro_star_value': 17.598674946465213}

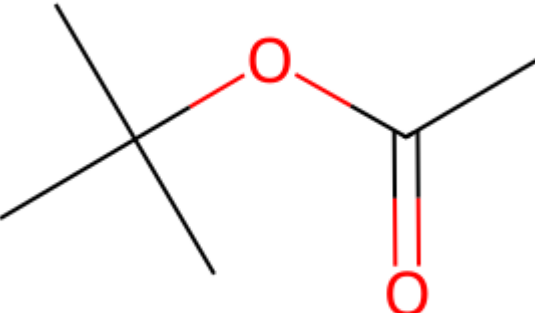


{'analysis_time': 0.601489,
'first_solution_time': 0.716983,
'num_calls_rxn_model': 5,
'num_calls_value_function': 158,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.44870909519674496,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 0.44870909519674496,
'retro_star_value': 17.598674946465213}

Reaction
{'analysis_time': 0.948311,
'first_solution_time': 0.950405,
'num_calls_rxn_model': 8,
'num_calls_value_function': 302,
'retro_star_min_cost': 11.52297580082373,
'retro_star_reaction_number': 11.52297580082373,
'retro_star_rxn_cost': 0.01005033585350145,
'retro_star_value': 18.464849400094295}

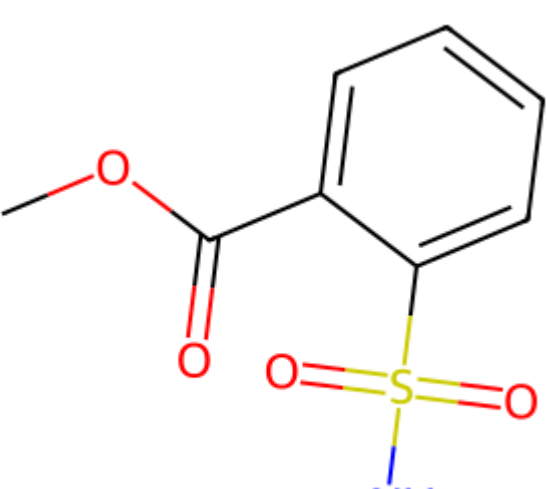


{'analysis_time': 0.948311,
'first_solution_time': 0.950405,
'num_calls_rxn_model': 8,
'num_calls_value_function': 302,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 11.512925464970229,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 11.512925464970229,
'retro_star_value': 18.464849400094295}

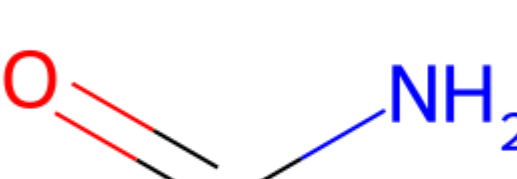


{'analysis_time': 0.716983,
'first_solution_time': 0.716983,
'num_calls_rxn_model': 6,
'num_calls_value_function': 241,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 22.928300273565846}

Reaction
{'analysis_time': 0.950405,
'first_solution_time': 0.950405,
'num_calls_rxn_model': 8,
'num_calls_value_function': 334,
'retro_star_min_cost': 11.512925464970229,
'retro_star_reaction_number': 11.512925464970229,
'retro_star_rxn_cost': 11.512925464970229,
'retro_star_value': 18.464849400094295}



{'analysis_time': 0.950405,
'first_solution_time': 0.950405,
'num_calls_rxn_model': 8,
'num_calls_value_function': 334,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 18.464849400094295}



{'analysis_time': 0.950405,
'first_solution_time': 0.950405,
'num_calls_rxn_model': 8,
'num_calls_value_function': 334,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 18.464849400094295}